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% You work for a new startup recording company as an audio engineer. One of
% your first record deals is a hot new Electronic Dance Music DJ that has
% recently switched to your label.
% There have been several issues with your company and you fear that the DJ
% may want to leave. Unfortunately, during the last recording session there
% was significant noise in the audio signal that you captured.
% As an audio engineer, you may be able to remove the noise from the signal
% if it does not ruin the desired audio. You have the audio clip named
% "lab_12_audio.wav" which contains the recorded data. In order to keep your
% biggest client, you must denoise this signal.
% Are you able to remove the noise from this audio signal?
% NOTE: Use "audioread" to import the noisy audio signal.
% Using the "filterDesigner", you must generate filter coefficients. Use the
% following constraints when you design your filter:
% 1. Order = 80 (81 coefficients will be generated)
% 2. Density factor = 20
% 3. Equiripple FIR
% 4. Transition Bandwidth = 0.05 in normalized units (|wstop-
% wpass|=0.05)
% 5. Stopband amplitude should be below -60 dB.
% 6. Export coefficients named "bb" to the mat file "<lastname>_bb.mat"
% Your script should load the "<lastname>_bb.mat" file you generated into
% the
% workspace before filtering the audio data. You can use either "conv" or
% "filter" command to filter the audio data.
% Plot your resulting spectrogram in Figure 1 with 1024 segments (each fft
% should be 1024 samples long).

[x, Fs] = audioread("lab_12\lab_12_audio.wav");
soundsc(x, Fs);

spectrogram(x, 512, 0);
m = conv(x, Num);
spectrogram(m, 1024, 0);

soundsc(m, Fs);
```

